Department of Electrical, Electronic and Computer Engineering

**Assessment ID: 2024EBN111A06**



**Electricity and Electronics EBN 111**

Release date: 17 April 2024

| | | | |
|---|---|---|---|
| Marks: | 15 | Total number of pages: | 8 |
| Submission Deadline: | 26 April 2024 (23:59) | Open/Closed book: | Open |

## Important instructions

1. This PDF file must be used in together with the **Excel answer spreadsheet** found on your AMS system account. The **Excel answer spreadsheet** offers each of you the unique parameter values for each question. Some extra instructions are also available on the **Excel answer spreadsheet**.
2. Final answers must be written as real numbers and not as fractions. (Use at least 5 significant figures to write irrational numbers.)
3. Answers must include units where applicable. (Refer to the EECE Rules for Electronic Assessments of tests and examinations for guidelines on how to enter units. Also, refer to the example list of units on the **Excel answer spreadsheet**.)
4. You can upload the **Excel answer spreadsheet** to AMS multiple times before the submission deadline. You can self-grade two attempts of these uploads to check and possibly improve your answers. The mark for each assignment is awarded according to the final submission of the **Excel answer spreadsheet**.

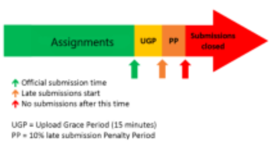
**Lecturers:** Prof. X. Ye, Mr. J. Thiselton, Mr. L. Staphorst

**Assistant Lecturers:** J. Nepembe, J. Masela, S. Twala, L. Hurter

**Online assessment submission rules and time-line**

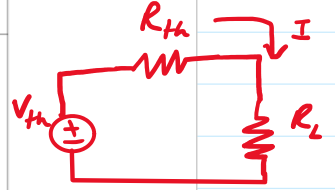
The following rules and time-line apply for all Assignments for EBN111/122 as illustrated by the figure below:

1. The start of the green section in the figure indicates the release of an assignment. The release of the assignment will be announced on ClickUP.
2. The official due date for the assignment is indicated by the green arrow in the figure. The due dates are also shown on the Lecture Schedule on ClickUP and on the AMS.
3. You must submit your **Excel answer spreadsheet** with your answers to each assignment on the AMS before the due date (green arrow in the figure below).
4. If you experience a technical issue with respect to uploading your assignment on the AMS, you can request assistance via **e-mail** ebn111.2024@gmail.com. Note, that except for extraordinary cases, you cannot submit your answers via email.
5. The submission due date will be followed by an Upload Grace Period of 15 minutes during which you must submit your answers on the AMS even if not finished with the assignment. No email submissions will be accepted during this period.
6. The Upload Grace Period is followed by a Penalty Period of 10 minutes. If you submit during this period, a 10% penalty will be applied to your final mark.
7. No late submissions will be accepted after the Penalty Period.

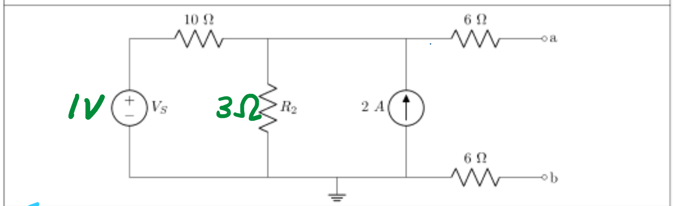


# Thevenin Theorem

Simplify circuit to:



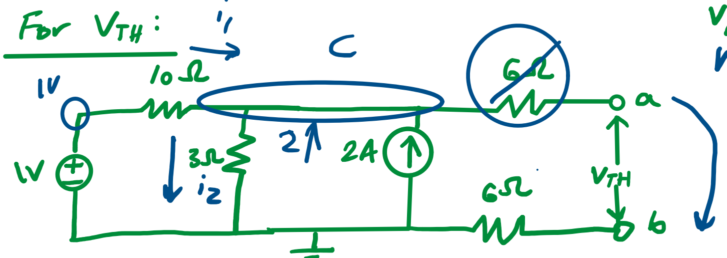

| Questions 1 to 3 [4] | |
|---|---|
| Given $V_S$ and $R_2$. Find the Thevenin equivalent voltage $V_{TH}$ and resistance $R_{TH}$, and then calculate the maximum power that can be transferred to a load resistor $R_L$ connected between terminals a-b. | |




| | |
|---|---|
| 01 | $V_{TH}$ [2] |
| 02 | $R_{TH}$ [1] |
| 03 | $P_{max}$ [1] |

$\boxed{01}$

For $V_{TH}$:




$V_B = 0V$

$V_C = V_A = V_{TH}$

No current

Nodal analysis at C:

$i_1 - i_2 + 2 = 0$

$\frac{1V - V_c}{10} - \frac{V_c}{3} + 2 = 0$

$V_c = 4.846153846\ V$

$\therefore V_{TH} = 4.846153846\ V$

For $P_{max}$:

$R_L = R_{TH}$

$P_{max} = \frac{\frac{1}{4} \times V_{TH}^2}{R_{TH}}$

$= \frac{\frac{1}{4} \times (4.846..)^2}{(14.3076...)}$

$= 0.4103598014W$

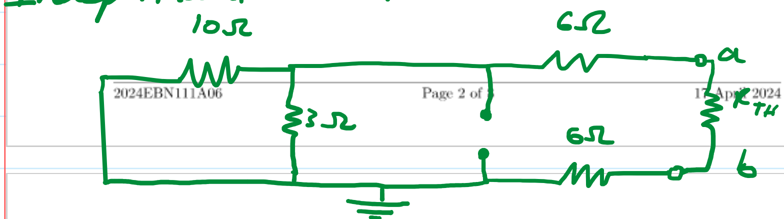
For $R_{TH}$:

Independant VS: short

Independant CS: open

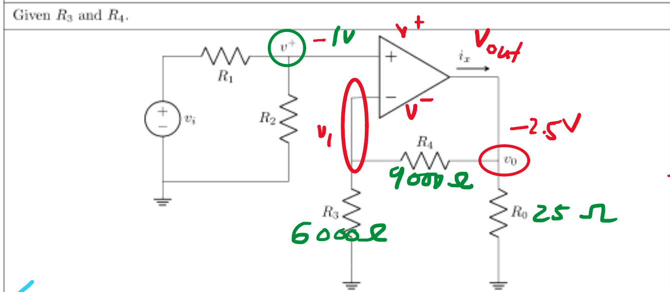
$R_{th} = \left(\frac{1}{10} + \frac{1}{3}\right)^{-1} + 6 + 6$

$= 14.30769231\Omega$






| Questions 4 to 5 [2] | |
|---|---|
| Given $R_3$ and $R_4$. | |




| | |
|---|---|
| 04 | Find the current $i_x$ when $R_0 = [R_0]$ Ω and $v^+ = -[V+]$ V. [1] |
| 05 | Calculate the closed-loop gain $(v_0/v_i)$ of the circuit when $R_1 = R_2$. [1] |

$V_1 = V^+ = -1V$

$\boxed{04}$

@ $V_1$

$\frac{-V_1}{6000} + \frac{-V_1 - V_o}{9000} = 0$

$V_o = -25V$

@ $V_o$:

$\frac{V_o}{25} + \frac{V_o + V_1}{9000} - i_x = 0$

$i_x = -0.10016666\ A$

$= -100.1666\ mA$

$\boxed{05}$

@ $V^+$

$\frac{V^+ - V_i}{R_1} + \frac{V^+}{R_2} = 0$

$\frac{-1 - V_i}{R_1} + \frac{-1}{R_1} = 0$

$$\frac{\quad}{R_1} + \frac{\quad}{R_1} = 0$$

$$-1 - V_i - 1 = 0$$

$$-V_i = 2$$

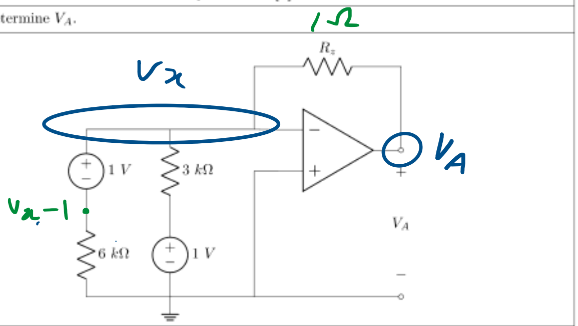
$$V_i = -2\,V$$

$$V_o / V_i = -2.5/-2 = 1.25$$



**Question 6 [1]**

Given $R_x$, determine $V_A$.




06 $V_A$ [1]

KCL @ $V_x$:

$$\frac{V_x - 1}{6000} + \frac{V_x - 1}{3000} + \frac{V_x - V_a}{1} = 0$$

$$V_x = V_a$$

$$\frac{V_a - 1}{6000} + \frac{V_a - 1}{3000} + \frac{V_a - V_a}{1} = 0$$

$$V_a = 1V$$

WRONG!

$$\frac{0-1}{6000} + \frac{(0-1)}{3000} + \frac{(0 - V_a)}{1\Omega} = 0$$

$$V_x = 0$$

$$V_A \neq V_x$$

$$0 - 1 + -2 \quad - 6000 V_a = 0$$

$$-3$$

$$V_a = \frac{-1}{-6000} = \frac{-1}{-3000}$$

$$\boxed{-5 \times 10^{-4}}$$

$$\frac{V_x - 1}{6000} + \frac{V_x - 1}{3000} + \frac{V_x - V_A}{1} = 0$$

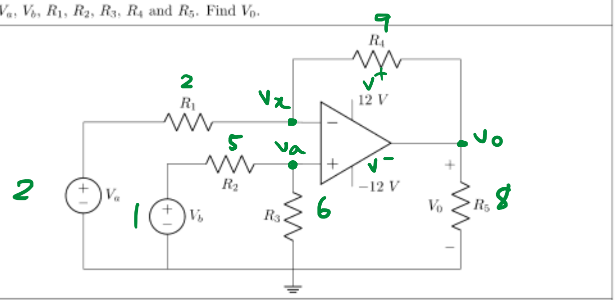
$$\frac{0-1}{6000} + \frac{0-1}{3000} + \frac{0 - V_A}{1} = 0$$

$$V_A = -5 \times 10^{-4}\,V$$

$$V_A = -500 \times 10^{-6}\,V$$

$$\boxed{V_A = -500\,\mu V}$$

| **Question 7** [1] |
| --- |
| Given $V_a$, $V_b$, $R_1$, $R_2$, $R_3$, $R_4$ and $R_5$. Find $V_0$. |




| | |
| --- | --- |
| Q7 | $V_0$ [1] |

@ $V_a$:

$$\frac{V_a - 1}{5} + \frac{V_a}{6} = 0$$

$$V_a = 0.545454 \text{ V}$$

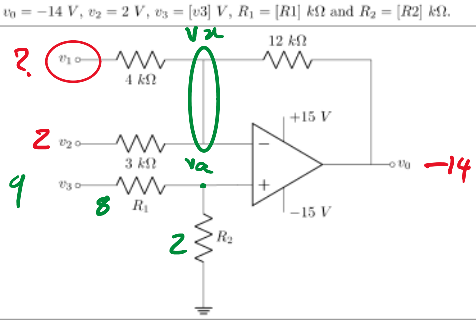
@ $V_x$:

$$\frac{V_x - 2}{2} + \frac{V_x - V_0}{9} = 0$$

$$V_a = V_x$$

$$\therefore V_0 = -6\text{V}$$

| Question 8 [1] |
| --- |
| Find $v_1$ when $v_0 = -14\ V$, $v_2 = 2\ V$, $v_3 = [v3]\ V$, $R_1 = [R1]\ k\Omega$ and $R_2 = [R2]\ k\Omega$. |




| | |
| --- | --- |
| 08 | $v_1$ [1] |

@$V_a$:

$$\frac{V_a - 9}{8} + \frac{V_a}{2} = 0$$

$$V_a = 1.8\ V$$

@$V_x$:

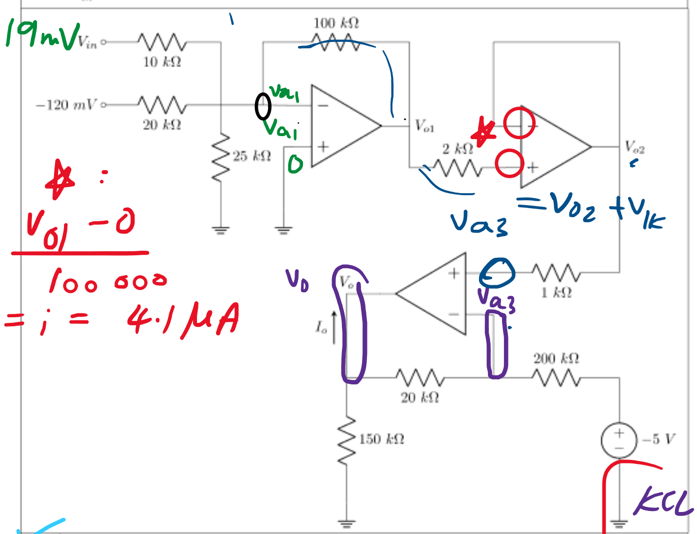
$$\frac{V_x - 2}{3000} + \frac{V_x - V_1}{4000} + \frac{V_x - V_o}{12000} = 0$$

$$V_x = V_a = 1.8\ V$$

$$\therefore V_1 = 6.8\ V$$

**Questions 9 to 11 [4]** $\boxed{9}$

Given $V_{in}$.




| | |
|---|---|
| 09 | Determine $V_{o1}$ [1] |
| 10 | Determine $V_{o2}$ [1] |
| 11 | Assuming $V_{o2} = \lvert V_{o2}\rvert$ $V$, determine $V_o$ [2] |

$V_{x_1} = V_{a_1} = 0V$

@$V_{x_1}$ :

$$\frac{V_{x_1} - 19\times10^{-3}}{10\,000} + \frac{V_{x_1} - -120\times10^{-3}}{20\,000} + \frac{V_{x_1}}{25000} + \frac{V_{x_1} - V_{o_1}}{100\,000} = 0$$

$V_{o_1} = 0.41\ V$

$\boxed{10}$ ~~WRONG BUT~~ why? :

$V_{01} = V_{02} = 0.41V$

~~MARKED RIGHT~~

$\boxed{11}$

$V_{a_3} = V_{o2} = 1V$

KCL @$V_{a_3}$ :

$$\frac{V_{a_3} - 5}{200\,000} + \frac{V_{a_3} - V_o}{20\,000} = 0$$

$$\frac{1+5}{200\,000} + \frac{1 - V_o}{20\,000} = 0$$

$V_o = 1.6\ V$

RIGHT method

$*$ :

$$\frac{V_{01} - 0}{100\,000} = i = 4.1\mu A$$

$V_{a_3} \neq V_{o2}$

$V_{a3} = V_{o2} + V_{1k}$

$= .1V + 1000i$

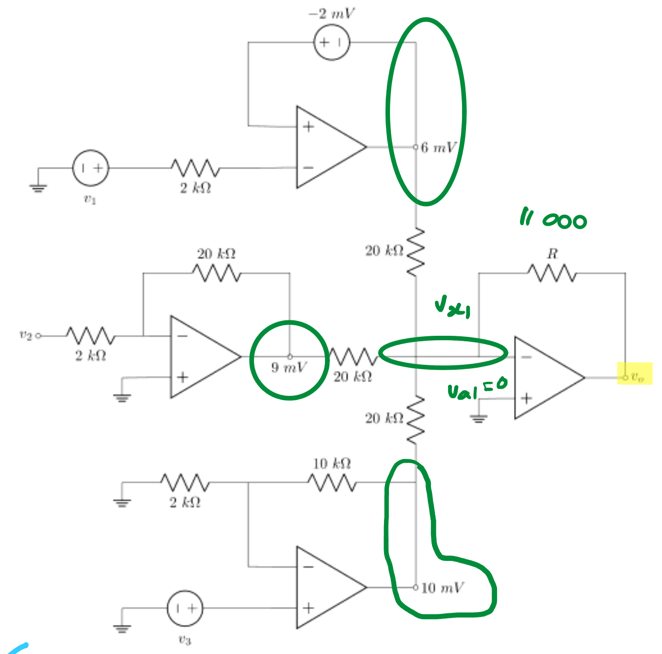
$= 1 + 1000(4.1\times10^{-6})$

$= 1.0041\ V$

WRONG!

**Question 12 [2]**

Given R, calculate $v_o$




| 12 | $V_o$ [2] |
|---|---|

$V_{a1} = V_{x1} = 0V$

KVL @ $V_{x1}$ :

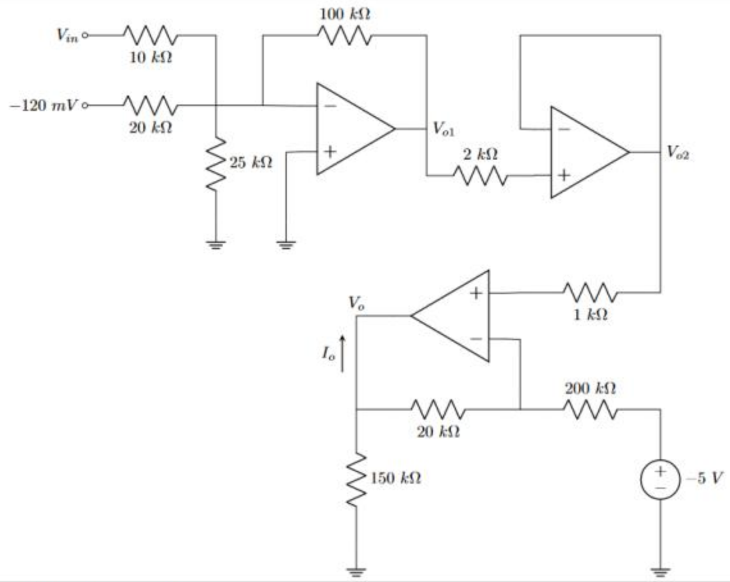
$$\frac{V_{x1} - V_o}{11\ 000} + \frac{V_{x1} - 6\,mV}{20\ 000} + \frac{V_{x1} - 9\,mV}{20\ 000} + \frac{V_{x1} - 10\,mV}{20\ 000} = 0$$

$$V_o = -0.01375\ V$$
$$= -13.75\ mV$$

**Grand Total : [15]**

**Questions 9 to 11 [4]**

Given $V_{in}$.




| | |
|---|---|
| **09** | Determine $V_{o1}$ [1] |
| **10** | Determine $V_{o2}$ [1] |
| **11** | Assuming $V_{o2} = [Vo2]\ V$, determine $V_o$ [2] |